🧠 Enterprise-Level MCQs

1.

You are reading a large CSV (5GB) using Pandas, but the system crashes due to memory error. What is the BEST approach?

A. Use `read_csv()` normally
B. Use `chunksize` in `read_csv()`
C. Convert CSV to JSON
D. Increase recursion limit

Answer: B

2.

A function modifies a list passed as an argument, impacting the original list. What is the reason?

A. Lists are immutable
B. Python uses pass-by-reference (object reference)
C. Function creates deep copy automatically
D. Lists are stored in stack

Answer:B

3.

You want to ensure a block of code runs regardless of exception occurrence. Which block is used?

A. except
B. else
C. finally
D. assert

Answer:C

4.

You need to validate email format in user input efficiently. Best approach?

A. String split
B. Regular Expression
C. Loop through characters
D. JSON validation

Answer:B

5.

In a REST API, which HTTP method is **idempotent**?

A. POST
B. PATCH
C. PUT
D. CONNECT

Answer:C

6.

A Flask API is slow under concurrent requests. What is the BEST improvement?

A. Add print statements
B. Use multithreading or deploy with Gunicorn
C. Use global variables
D. Disable logging

Answer:B

7.

You want to restrict access to certain API routes in FastAPI. What should you use?

A. Decorators with dependency injection
B. Only comments
C. Print statements
D. JSON schema

Answer:A

8.

Which scenario BEST justifies using a generator?

A. Small dataset processing
B. Large streaming data
C. Static data storage
D. Recursive functions

Answer:B

9.

In multithreading, race conditions occur due to:

A. Parallel execution without synchronization
B. Slow CPU
C. Memory overflow
D. Syntax errors

Answer:A

10.

Which decorator is used to measure execution time of functions?

A. @staticmethod
B. Custom decorator
C. @classmethod
D. @property

Answer:A

11.

You need to ensure a database transaction either completes fully or rolls back. Which concept applies?

A. Polymorphism
B. ACID properties
C. Encapsulation
D. Recursion

Answer:B

12.

In SQLAlchemy, ORM helps in:

A. Writing raw SQL faster
B. Mapping Python objects to database tables
C. Avoiding database usage
D. Converting JSON to CSV

Answer:B

13.

A Docker container is removed after stopping. What is missing?

A. Image
B. Volume
C. Restart policy
D. Network

Answer:C

14.

You want environment-specific configs (dev/prod). Best approach?

A. Hardcode values
B. Use environment variables
C. Use print statements
D. Use recursion

Answer:B

15.

In Git, which command safely integrates changes from remote without overwriting local work?

A. git push
B. git clone
C. git pull
D. git init

Answer:C

16.

In Pytest, reusable setup code is implemented using:

A. Decorators
B. Fixtures
C. Classes
D. Lambda

Answer:B

17.

You want to validate incoming API request data in FastAPI. What should you use?

A. Dictionaries
B. Pydantic models
C. JSON files
D. Regex only

Answer:B

18.

A recursive function runs infinitely. What is missing?

A. Loop
B. Base condition
C. Lambda
D. Exception

Answer:B

19.

While reading a file, you want automatic closure even if error occurs. Use:

A. open()
B. try only
C. with statement
D. lambda

Answer:C

20.

You are designing a scalable API. Which principle should you prioritize?

A. Tight coupling
B. Stateless design
C. Hardcoding
D. Global variables

Answer:B